

Module 2 — Read 2

z-scores, statistics toolbox, and one numerical variable

Module 2, Read 2: z-scores, toolbox, and one numerical variable

Project: [Project 2 — Estimating the Truth](#) · **Prerequisite:** [Read 1](#) and [Quiz A](#) · **Next:** [Quiz B](#), demo lab, exercise lab.

Complete Read 1 and Quiz A first. This reading adds **standardization**, the **methods-map habit**, and **summaries/graphs** for one numerical variable. `t.test()` for hypothesis testing is **optional** for Project 2 (CI focus).

Learning outcomes

Three sections — use **Check yourself** before [Quiz B](#).

After this reading you should be able to:

Section 1: Normal distribution and standardization

Normal distribution and standardization

- Compute a **z-score** from an observation and the **mean** and **standard deviation** of a reference distribution.
- **Interpret** a z-score in context and use z-scores for **standardized comparisons** between groups or to a reference.
- Find **probabilities** (areas) for the **standard normal** distribution and describe them as proportions of values below (or above) a cutoff.

Section 2: Organizing the statistics toolbox

Organizing the statistics toolbox — z-test for a proportion, variable types, CI vs. testing

- Compute a **standardized test statistic** using the **normal approximation to the binomial** and connect it to a **one-proportion** test (often called a **z-test**); in R, use `prop.test()` and read off a **confidence interval** as well.
- Organize analyses by **type of variable(s)** and by **purpose** (e.g. description vs. inference).
- Relate **descriptive** tools to **inferential** choices for **one categorical** variable (e.g. proportion, confidence interval vs. hypothesis test).

Section 3: One numerical variable — center, graphs, and t-tools

One numerical variable — center, graphs, and the one-sample t-test

- Compute **mean** and **median** by hand and with the **tidyverse** (pipe to `summarize()`).
- Graph one numerical variable with **ggplot2** using a **histogram** and a **dot plot**.
- State the **one-sample t-statistic** and use `t.test()` for a **one-sample t-test** and **confidence interval** for the **population mean**.

Section 1: Normal distribution and standardization

Normal distribution and standardization

Bridge from Read 1

You already used the **Empirical Rule** and normal curves for binomial counts in **Read 1**. Here we **standardize** any normal reference to compare observations on a common scale (z-scores) and read off areas with the **standard normal**.

In this section

Example: Who is taller, Rosanna or Peter?

Two observations:

Annarose's height: **67** inches (women's reference group).

Peter's height: **72** inches (men's reference group).

So **Peter** is taller when heights are compared directly (72 vs 67 inches).

But who is taller relative to their group?

Draw a reference distribution for the stature of men and a reference distribution for the stature of women.

Identify the location of the center of the stature distribution on each plot and show one and two standard deviations (SD) on either side.

Add vertical lines to the appropriate distribution at Rosanna and Peter's heights. Who appears to be taller compared with the reference distributions?

According to the plots, **Annarose** is taller relative to her group (see z-scores below).

Instead of plotting two different reference distribution and comparing the observations where they fall relative to two SD from the center, we can compare the heights relative to their groups directly via **z-scores**.

Definition: z-score

$$z = \frac{\text{observation} - \text{mean of reference}}{\text{SD of reference}}$$

A z-score measures the number of SDs the observation is above the mean (or below if the difference is negative).

Women: $\mu = 64$, $\sigma = 2.5$. Annarose 67 in $\rightarrow z = (67 - 64)/2.5 = 1.2$.

Men: $\mu = 70$, $\sigma = 3$. Peter 72 in $\rightarrow z = (72 - 70)/3 \approx 0.67$.

Annarose has the larger z-score — she is more unusually tall **for her group**.

If we assume the shape of a reference distribution is *Normal* then we can use a table or online calculator or R to find the percentage in the population that corresponds to a particular z-score.

Definition: Standard Normal

A standard normal distribution is a normal distribution with mean 0 and SD 1.

Notice that by the Empirical Rule, we can see that 68% of values are between -1 and 1, 95% are between -2 and 2, and 99% are between -3 and 3. The proportions of values corresponding to all those below a particular value are given in a **Standard Normal Table**:

[normal-table.pdf](#)

[Online Calculator](#)

In R, `pnorm(z)` returns the area **to the left of** z on the standard normal curve:

```
pnorm(1.24) # proportion at or below z = 1.24
pnorm(-1.94) # grip-strength example (~0.03)
```

Example: Rosanna's grip strength this morning using her dominant hand was 38.8 lbs. Suppose that mean grip strength is 66lbs and SD is 14lbs for women in their early forties in the US. Compute Rosanna's z-score and use it to find the percentage of women in their early forties in the US that Rosanna is stronger than assuming that this distribution is normal.

i Worked solution — grip strength

$z = (38.8 - 66)/14 \approx -1.94$. About **3%** of the reference group has grip strength **below** hers (use the table, [calculator](#), or `pnorm(-1.94)` in R).

Optional: measure your own grip strength, pick a reference row from the table below, and compute $z = (\text{yours} - \mu)/\sigma$.

Group	Mean (lbs)	SD (lbs)
College aged men	104	18
College aged women	62	16
Great Grandmothers	44	10

Use the normal table or calculator: $P(Z < z)$ is the proportion **weaker than** you when z is your score.

(use the online calculator or Standard Normal Table) ::: {.callout-tip icon=false} ## Check yourself — Normal distribution and standardization

1. Women's heights: $\mu = 64$, $\sigma = 2.5$ inches. Annarose is 67 inches. Compute her z-score.
2. Men: $\mu = 70$, $\sigma = 3$. Newton is 72 inches. Compute his z-score.
3. Who is more unusually tall **relative to their group**, Annarose or Newton?
4. If Z is standard normal, what does $P(Z \leq 1.24)$ represent?

:::

Answers — Normal distribution and standardization

1. $z = (67 - 64)/2.5 = 1.2$ — she is 1.2 SD above the women's mean.
 2. $z = (72 - 70)/3 \approx 0.67$.
 3. Annarose ($z = 1.2$ vs $z = 0.67$).
 4. The proportion of standard-normal values **at or below** 1.24 (about 89%).
-

Section 2: Organizing the statistics toolbox

Organizing the statistics toolbox

In this section

- Compute a test statistic using the normal approximation to the binomial and use it to perform a test (often called a **z-test**); in R, `prop.test()` in one line also gives a confidence interval.
 - Use organizational tools: what type of data you have and the purpose of the analysis.
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Variables and tools (overview)

- **Variables** — types of variables
 - **Descriptive statistics** — by number and type of variable
 - **Inferential statistics** for one binary variable: confidence interval vs. testing
-

Standardizing statistics used to conduct a hypothesis test

Last lecture we saw the **z-score** as a way to compare relative standing in two different groups or within a group, assuming the reference group is normally distributed.

Let's apply the same idea of shifting and scaling to conducting a hypothesis test of a population proportion or probability p .

If $np \geq 10$ and $n(1 - p) \geq 10$, then the normal distribution can be used to approximate the **binomial** distribution of X = number of successes in n trials, where p is the probability of success in a single trial and the trials are independent.

In **Module 2 Read 1** you used this fact along with the empirical rule that 95% of the values of a normal distribution are within 2 SDs of the mean. We computed the mean of X as $\mu = np$ and the SD of X as $\sigma = \sqrt{np(1 - p)}$, and then rejected the null hypothesis if our observed value x was below $\mu - 2\sigma$ or above $\mu + 2\sigma$.

Another way to conduct this test is to **standardize** the observed value x by subtracting the mean and dividing by its standard deviation (also called **standard error** in this context):

Definition: standardized test statistic

$$\frac{\text{observed statistic} - \text{mean of sampling distribution}}{\text{standard deviation of sampling distribution}}$$

In the case of the statistic being a count of successes X from a binomial random process with n and p , the standardized test statistic is

$$z = \frac{x - np}{\sqrt{np(1 - p)}}.$$

This value can be compared with the standard normal distribution; a null hypothesis is rejected if z is below -2 or above 2 (the standard normal distribution has mean 0 and SD 1).

Equivalently, a standardized test statistic can be computed using the observed proportion $\hat{p} = x/n$: it is the same as the one based on x , but all terms are divided by n :

$$z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

Note that p is the value of the population proportion or probability hypothesized under the null hypothesis, and $\hat{p}$ is the observed proportion from n trials.

Example: Test whether a coin is fair if 16 heads were observed in 20 flips using a significance level of 0.05. Use a standardized test statistic and assume it follows the standard normal distribution.

i By hand first

Under $H_0 : p = 0.5$, $np = 10$ and $n(1-p) = 10$. $z = \frac{16 - 20(0.5)}{\sqrt{20(0.5)(0.5)}} = \frac{6}{\sqrt{5}} \approx 2.68$. This is beyond ± 2 , so we **reject** H_0 at $\alpha = 0.05$.

Same example in one line of R (optional for Project 2 — the project uses **CI**, not hypothesis test):

```
prop.test(x = 16, n = 20, p = 0.5)
```

1-sample proportions test with continuity correction

```
data: 16 out of 20, null probability 0.5
X-squared = 6.05, df = 1, p-value = 0.01391
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.5573138 0.9338938
sample estimates:
 p
0.8
```

Interpretation: By default the test is **two-sided** (the alternative is that the coin is not fair). The output includes not only the p-value but also the lower and upper bounds of a confidence interval.

Organizing our statistics toolbox

We've now seen quite a few tools for analyzing data. Let's think about how to keep them organized. It's important to consider what sort of data we **have** and the purpose of the analysis — what we **want**.

Examples of what we might want

The purpose of a **descriptive statistics** analysis is to describe relevant information contained in a particular dataset.

If our data are binary outcomes (e.g. successes/failures, yes/no, good/bad), we have described them using:

- Counts
- Proportions

Example: Five students were asked: what is the maximum number of hot dogs you could eat?

Review notation: x , n , and $\hat{p} = x/n$ for count, sample size, and observed proportion.

If our data are a set of numbers, we summarize the **center** with the mean and the **spread** with a standard deviation.

In general, the type of calculation we use to describe a dataset depends on the *type* of information we have. Here are two useful definitions:

Definitions

A **variable** is a characteristic that may vary from object to object or person to person in our population of interest.

We'll consider two **types** of variables:

- **Categorical** (qualitative) — distance or order between values is not meaningful.
- **Numerical** (quantitative) — distance and order are meaningful.

Examples: Classify each of the following variables as categorical or numerical:

- Age of a student in years
- Year a student was born
- Number of hot dogs eaten at a picnic
- Favorite beach of a student
- Rating on a scale of 1–5 for how much a student enjoys a statistics class

Steps in a descriptive analysis

1. Think about the purpose of your description. Decide what is “relevant.”
2. Classify each variable as either categorical or numerical.
3. If the variable is numerical, summarize its center and spread using mean and SD (more options will be discussed in the next lecture).
4. If the variable is categorical, summarize it using counts and proportions.

Inferential tools

If we want to infer the value of a parameter in the population from sample data, we currently have two **inferential** tools: confidence intervals and tests.

Confidence intervals answer: “What is a range of values that are reasonable for the parameter?” **Tests** compare the parameter against a **hypothesized** value.

We already have tests and confidence intervals for a population proportion/probability p ; these apply when the data are binary categorical.

We'll spend the next few weeks expanding these tools to cover the cases of one numerical variable, two categorical variables, one numerical and one categorical variable, and two numerical variables.

We'll also see a **third** inferential tool: **prediction**. ::: {.callout-tip icon=false} ## Check yourself — Organizing the statistics toolbox

1. Write the standardized statistic for testing $H_0 : p = p_0$ using observed proportion $\hat{p}$.
2. Stem diameter (cm) for trees — numerical or categorical?
3. Parasite present: yes/no — numerical or categorical? Binary?
4. One binary variable from a sample: name one **descriptive** summary and one **inferential** tool.

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Answers — Organizing the statistics toolbox

1. $z = (\hat{p} - p_0) / \sqrt{p_0(1 - p_0)/n}$ when normal approx conditions hold.
 2. **Numerical** (measurement).
 3. **Categorical, binary**.
 4. Descriptive: $\hat{p}$ or count. Inferential: 95% CI for p or binomial/z test.
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Section 3: One numerical variable — center, graphs, and t-tools

One numerical variable — center, graphs, and t-tools

In this section

i Optional for Project 2

Project 2 may use a **mean** CI via $\bar{x} \pm 2s/\sqrt{n}$ (or `t.test` for small n). The one-sample **t-test** below is enrichment — not required on the Project 2 rubric.

- Compute **mean** and **median** by hand and with the **tidyverse** (pipe to `summarize()`).
- Graph one numerical variable with **ggplot2**: **histogram** and **dot plot**.
- Recognize the **one-sample t-statistic** and use `t.test()` for a **one-sample t-test** and **confidence interval** for the population mean.

Review: one numerical variable

A **numerical** variable takes values that are counts or measurements where it is meaningful to sort them in order and to compute distance between any two of them. We usually report center and spread; this lecture focuses on center and on graphs for the whole distribution.

Mean (average). For values $x_1, \dots, x_n$, the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The capital Greek letter Sigma (Σ) denotes **summation**. The expression $\sum_{i=1}^n x_i$ means: add $x_1 + x_2 + \dots + x_n$, including every observation of x , indexed by i from 1 to n . Multiplying that total by $\frac{1}{n}$ rescales it: you are taking the combined amount (the total sum) and expressing it as if it belonged to a single unit, “per person,” “per observation,” or “per employee” in a payroll setting.

How to read the mean. Think: total of all values, spread evenly across n slots. If everyone’s contribution were replaced by the same number while keeping the same total, that common number would be the mean.

Visual (balance point). On a **dot plot** with one dot per observation, the mean is the **balance point** (center of mass) along the horizontal axis: equal weight at each dot, and the axis would balance at $\bar{x}$. On a **histogram**, you get the same idea if you treat each bin’s count (or properly scaled density) as sitting at the **bin center**, that is a standard approximation; the exact mean always comes from the original values, not from the drawn bins.

Median. Sort the values from smallest to largest. If n is odd, the median is the middle value. If n is even, it is the average of the two middle values. In the usual case with no ties sitting exactly on the split, the median is the value such that half of the observations are below it and half are above it. (If many values tie at the median, the “50% / 50%” story is still right in spirit but the counts on each side need a careful definition, we keep the ordered-list rule for computing it.)

How to read the median. The median is a positional measure: it marks the center of the ordered list, not the balance of the total of the numbers. It is not pulled in dollars-and-cents by a few huge values in the same way the mean is.

Example: salaries. Suppose you list every employee’s annual salary. The **median** answers a “typical employee” question: half of the staff earn less than or equal to that amount and half earn more than or equal to it. The median

is a useful picture of how pay feels in the middle of the workforce. The **mean** answers a “budget” question: if the firm paid the same total in salaries but split it evenly across every employee, each would get the mean. So the mean is the per-employee share of the fixed payroll total; it is what you need if you want to think about resizing the organization (“what if we had twice as many people with the same total wage bill?”) or track aggregate cost per head.

Which should you report? There is no single correct choice for every problem. If the goal is to describe the feeling of a typical standing, the median is often easier to interpret. If the goal is to tie summaries to a total, a rate per unit, or planning with a fixed sum divided among n units, the mean is a better choice. Report what matches your question and when in doubt, both plus a graph can be clearer than either alone.

Examples by Hand

1. Data: 7, 8, 9, 10, 10, 11, 12, 13 .

- Mean: $\bar{x} = (7 + 8 + 9 + 10 + 10 + 11 + 12 + 13)/8 = 80/8 = 10$.
- Median: notice that the 8 values are already sorted from smallest to biggest and the two middle values are both 10, so the median is 10.

Here the mean and median match.

2. Data: 2, 3, 4, 5, 6, 7, 100 (seven values). One value (100) is far from the rest.

- Median: with seven values already sorted, the median is the fourth in the list so the median is 5.
- Mean: the sum is $2 + 3 + 4 + 5 + 6 + 7 + 100 = 127$, so $\bar{x} = 127/7 \approx 18.1$.

The mean is pulled toward the large observation; the median stays with the middle of the bulk of the data. For describing a “typical” case here, the median (5) is the point at which about half the values are less than it and half are greater than it. The mean reflects the total (127) spread across seven slots, including the one very large contribution and is the “balance point” of all seven values.

Mean and median in the tidyverse (pipe)

Syntax note: In **dplyr**, `summarize()` and `summarise()` are the same function (American vs. British spelling). For pipes, base R’s native `|>` (R 4.1+) works like the **magrittr** pipe `%>%` from the **tidyverse**: both pass the left-hand side into the first argument of the function on the right. This lecture uses `|>` and `summarize()`; use whichever spelling and pipe you prefer, they behave the same here.

The pipe `|>` sends the data on its left into the first argument of the function on its right. We often pipe a data frame (e.g. a dataset with examples in rows and variables in columns) into `summarize()` to compute summaries.

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

scores <- tibble(
  score = c(7, 8, 9, 10, 10, 11, 12, 13)
)

scores |>
  summarize(
    n = n(),
    mean_score = mean(score),
```

```
median_score = median(score)
)
```

```
# A tibble: 1 x 3
  n mean_score median_score
<int>    <dbl>    <dbl>
1     8      10      10
```

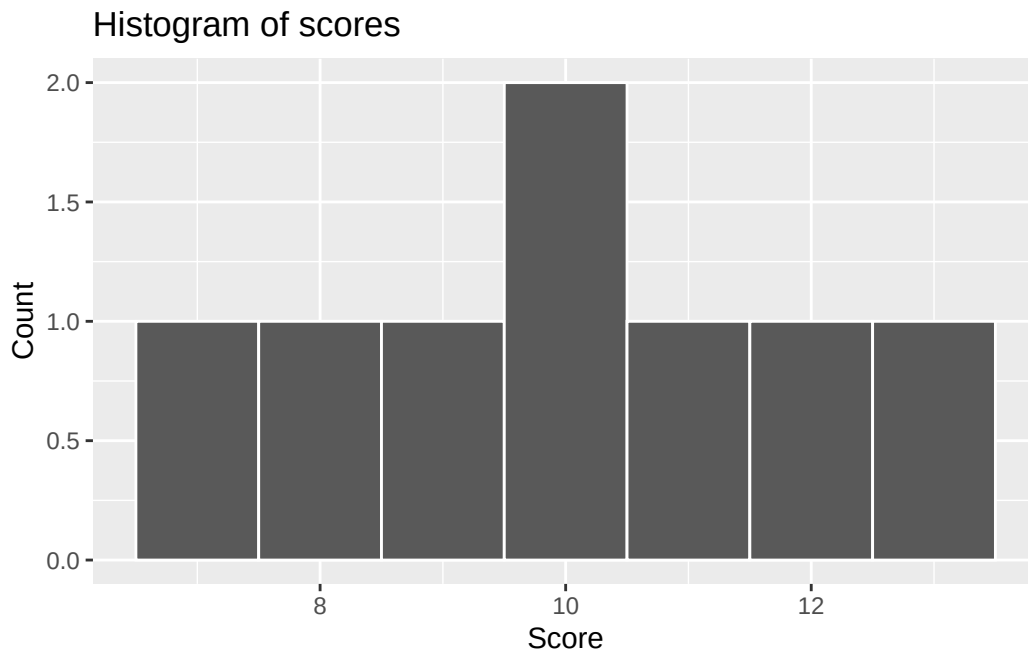
Here `n()` counts rows inside `summarize()` (one row per student, say).

Histogram and dot plot (ggplot2)

Histogram, split the axis into bins and show counts (or densities) in each bin.

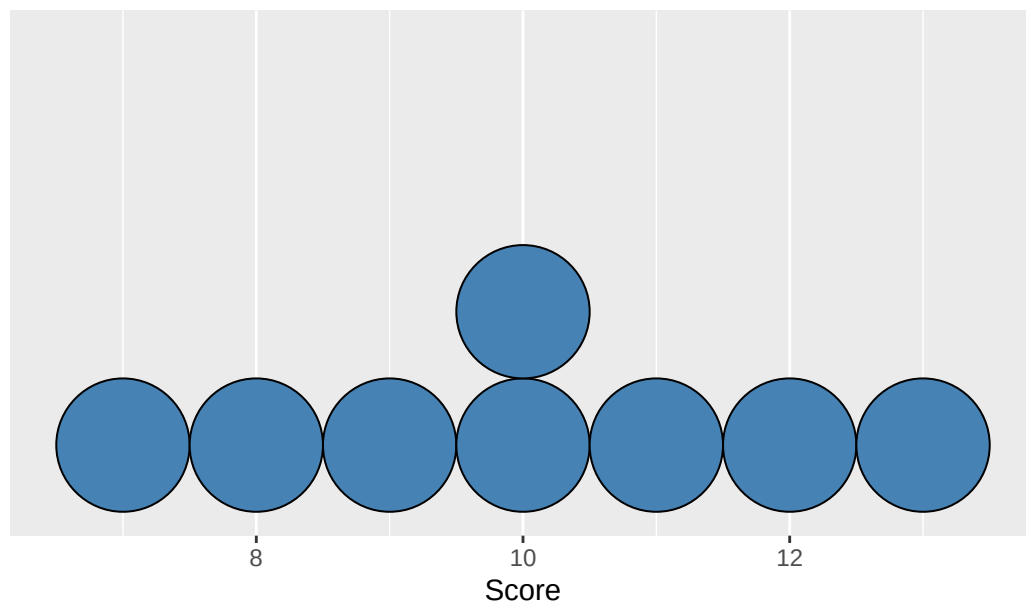
Dot plot, stack dots along the axis so you see each value (or rounded value) and repeats clearly.

```
ggplot(scores, aes(x = score)) +
  geom_histogram(binwidth = 1, boundary = 6.5, color = "white") +
  labs(
    title = "Histogram of scores",
    x = "Score",
    y = "Count"
  )
```



```
ggplot(scores, aes(x = score)) +
  geom_dotplot(binwidth = 1, method = "histodot", fill = "steelblue") +
  labs(
    title = "Dot plot of scores",
    x = "Score",
    y = NULL
  ) +
  scale_y_continuous(NULL, breaks = NULL) # hide stacking axis
```

Dot plot of scores



Adjust `binwidth` so the plot matches how you think about the scale (e.g. group by every 1 unit, 10 units, 100 units...) or leave this input out to let R use a default setting.

One-sample t -statistic and `t.test()`

Setting. We have a **simple random sample** of size n from a population with unknown mean μ . We want to test whether μ equals a hypothesized value μ_0 or to estimate reasonable values for μ via a **confidence interval**.

Definition: t -statistic (one sample). With sample mean $\bar{x}$ and sample SD s ,

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

The denominator is the **standard error** of the mean. Under $H_0 : \mu = \mu_0$, and when the usual conditions hold (random sample and either n is large (>30) or the population is roughly symmetric), t is compared to a **Student t distribution** with $df = n - 1$. The exact shape depends on sample size and is not calculable by hand. In modern times, we use software for p -values and intervals, not tables by hand.

`t.test()` for one sample: pass the numeric vector, the null mean `mu`, and optionally `alternative` ("`two.sided`", "`greater`", "`less`"). The default output includes a test and a 95% **confidence interval** for μ (two-sided alternative).

```
t.test(scores$score, mu = 9, alternative = "two.sided")
```

One Sample t-test

```
data: scores$score
t = 1.4142, df = 7, p-value = 0.2002
alternative hypothesis: true mean is not equal to 9
95 percent confidence interval:
 8.327958 11.672042
sample estimates:
mean of x
 10
```

Interpretation. The **confidence interval** is the set of estimates for μ that we believe contain the true value of μ with 95% confidence. The **p-value** summarizes evidence against $H_0 : \mu = \mu_0$ in favor of the stated H_a : reject the null H_0 in favor of H_a if the p-value is less than the significance level α (often taken to be 0.05 but remember you should pick this value based on the seriousness of a Type I error).

Takeaway

For a single **numerical** variable, **mean** and **median** describe center; **ggplot2** gives histograms and dot plots of the full distribution. For inference about one population mean, the **one-sample t-test** and interval come from `t.test()` on the sample vector. ::: {.callout-tip icon=false} ## Check yourself — One numerical variable — center, graphs, and t-tools

1. Data: 3, 5, 5, 7, 9. Compute mean and median.
2. Data: 2, 3, 4, 5, 6, 7, 100. Which summary is better for a “typical” salary?
3. In tidyverse, which function **adds a column** with the same number of rows?
4. Which ggplot geometry makes a **histogram** for one numerical variable?

:::

Answers — One numerical variable — center, graphs, and t-tools

1. Mean = 5.8. Median = 5.
2. **Median** (5) — mean is pulled by the outlier.
3. `mutate()`.
4. `geom_histogram()` (after `aes(x = variable)`).